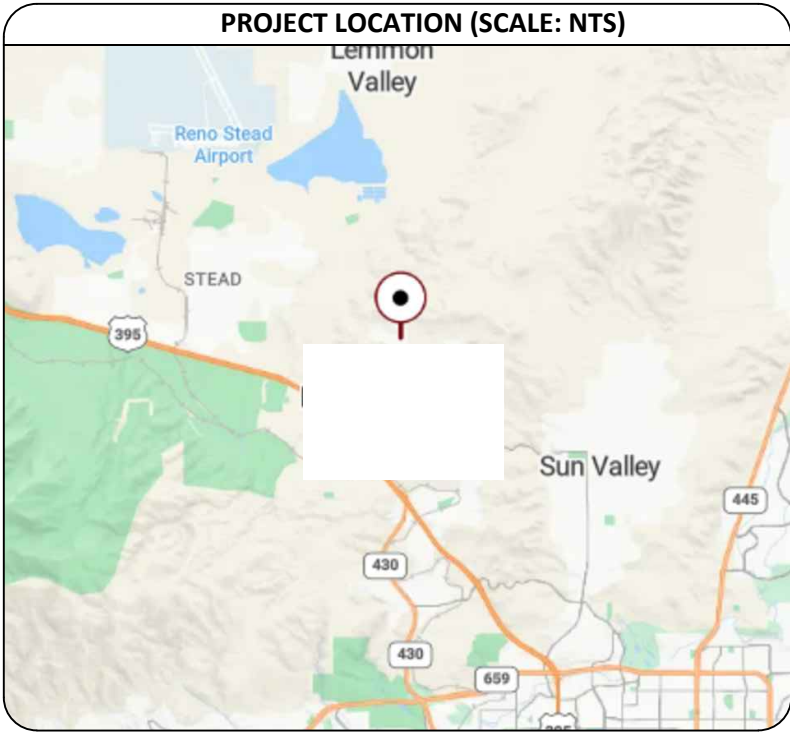
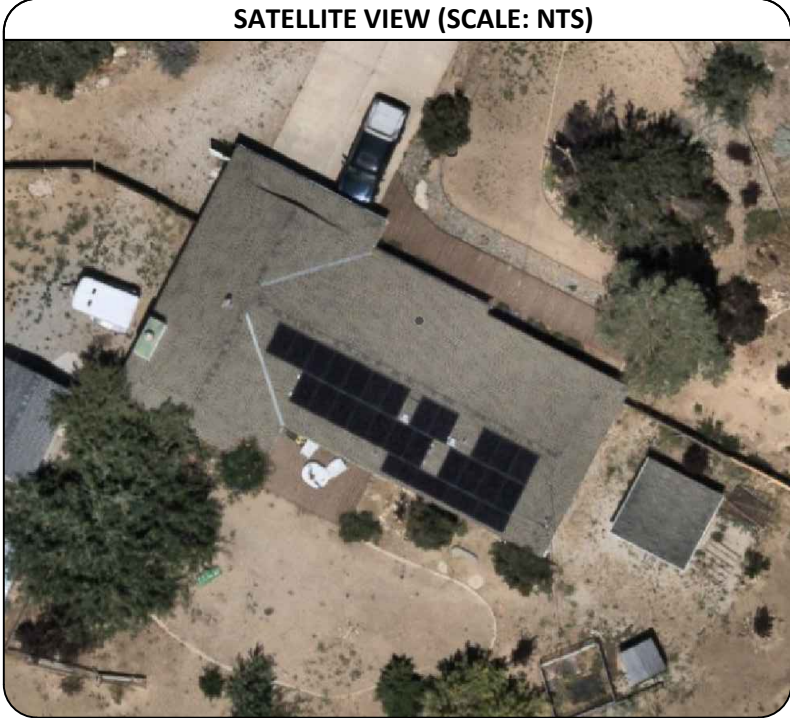


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(24) MODULES x 400W	(24) INVERTER x 349W
9.600 kW	8.376 kW



SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A RESIDENTIAL LOCATION: 1234 STREET NAME, RENO, NV 9874. COORDINATES: 39.622208, -119.828248 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM. - INSTALLED PV SYSTEM WILL BE CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ). - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

APPLICABLE GOVERNING CODES & NOTES
THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES: 2024 INTERNATIONAL BUILDING CODE 2024 INTERNATIONAL RESIDENTIAL CODE 2024 INTERNATIONAL MECHANICAL CODE 2024 INTERNATIONAL PLUMBING CODE 2024 INTERNATIONAL ENERGY CODE 2018 INTERNATIONAL FIRE CODE 2023 INTERNATIONAL ELECTRICAL CODE ALL OTHER RELATED STATE, LOCAL & AHJ AMENDMENTS.

DESIGN CRITERIA
- CONSTRUCTION: SINGLE FAMILY DWELLING (SFD) - NO. OF STORIES: SINGLE STORY - EXPOSURE CATEGORY : B, RISK CATEGORY : II - WIND SPEED (ASCE 7-22) : 96 MPH - GROUND SNOW LOAD (ASCE 7-22) : 72 lb/sqft - ASHRAE EXTREME LOW: -15°C - ASHRAE HIGH TEMP (2% AVG.) : 35°C - ASHRAE DISTANCE ABOVE ROOF (7/8") : 57°C

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
E-1	ELECTRICAL DIAGRAM & CALCULATION
E-2	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

PROJECT INFORMATION	
(#) PV MODULES	(24) QCELLS Q.PEAK DUO L-G5.2 400W
(#) MICRO - INVERTER	(24) ENPHASE IQ8A-72-2-US
(#) COMBINER BOX	(01) ENPHASE IQ COMBINER 4 (X-IQ-AM1-240-4)
SYSTEM OUTPUT CURRENT	34.80A
BRANCH CIRCUIT	03 CIRCUITS OF 08 MODULES
ROOF RACKING SYSTEM	UNIRAC FLASHKIT PRO W/ UNIRAC SOLARMOUNT LT
POINT OF INTERCONNECTION	50A LOAD SIDE BREAKER IN EXTERIOR NEW SOLAR READY MAIN SERVICE PANEL WITH 125A BUSBAR, 120/240V, 3W, 1φ
UTILITY AHJ	NV ENERGY WASHOE COUNTY

GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & NEC CERTIFIED, WHERE WARRANTED - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS. - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24. - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM. - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE. - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS. - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15. - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92. - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE. - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM. - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (NEC690.4(E) AND 705.6) - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF. - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES. - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL. - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS . - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES. - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT. - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO NEC. - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED. - ALL EXTERIOR CONDUIT TO BE PAINT MATCHED TO THE HOMES COLORS

CONTRACTOR INFO	DATE	08/15/2025
	PROJECT ID	SP-1
	CREATED BY	TV
	CHECKED BY	DS
	SCALE	NTS
	SHEET SIZE	11"X 17"
		ANSI B

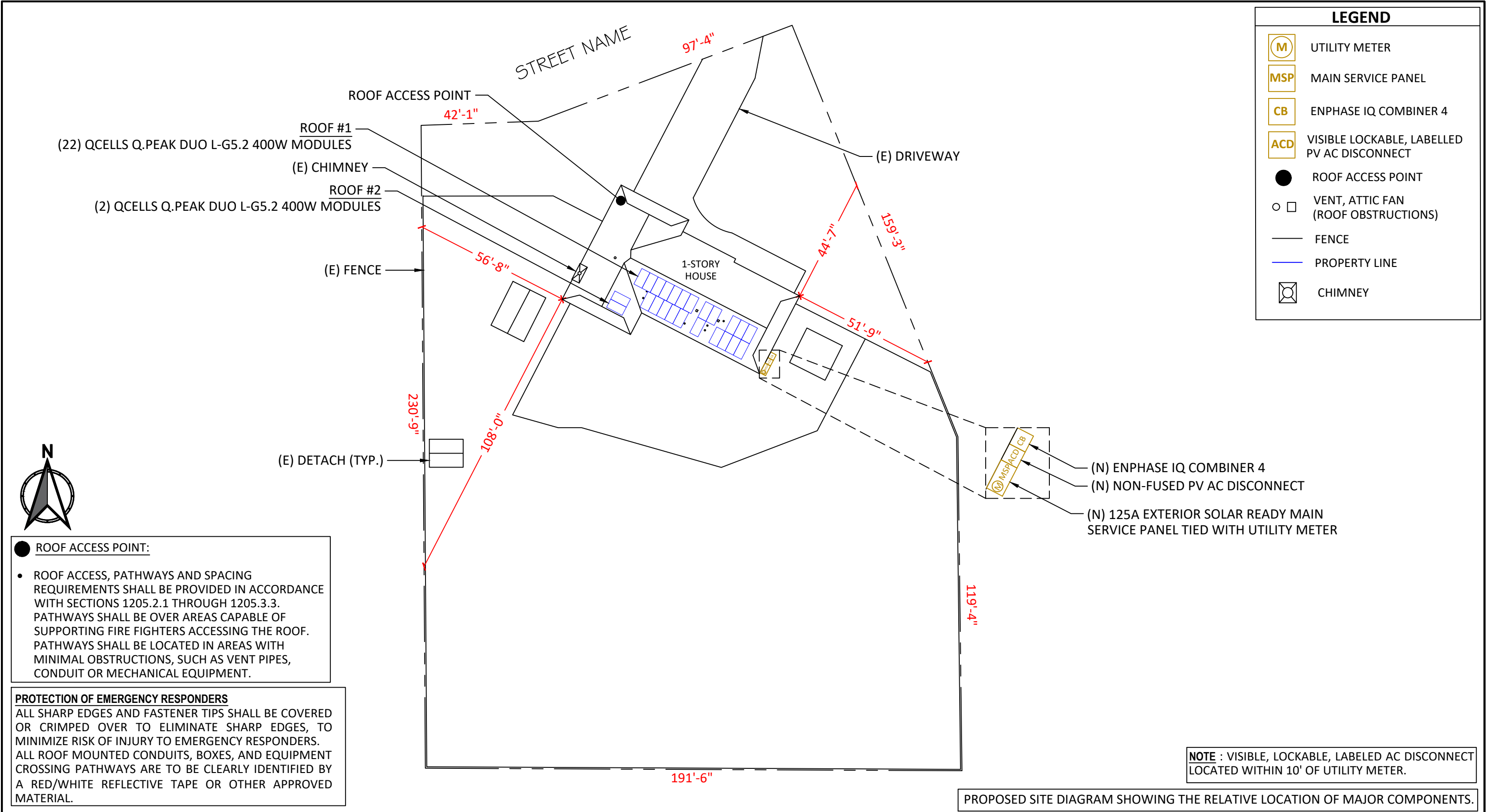
DATE	08/15/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	DS
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT NAME RESIDENCE	
1234 STREET NAME, RENO, NV 9874	
APN NUMBER: 147852369	
HOMEOWNER INFO	

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY	

PV-0
SHEET NO.
PROJECT SUMMARY
SHEET INFORMATION

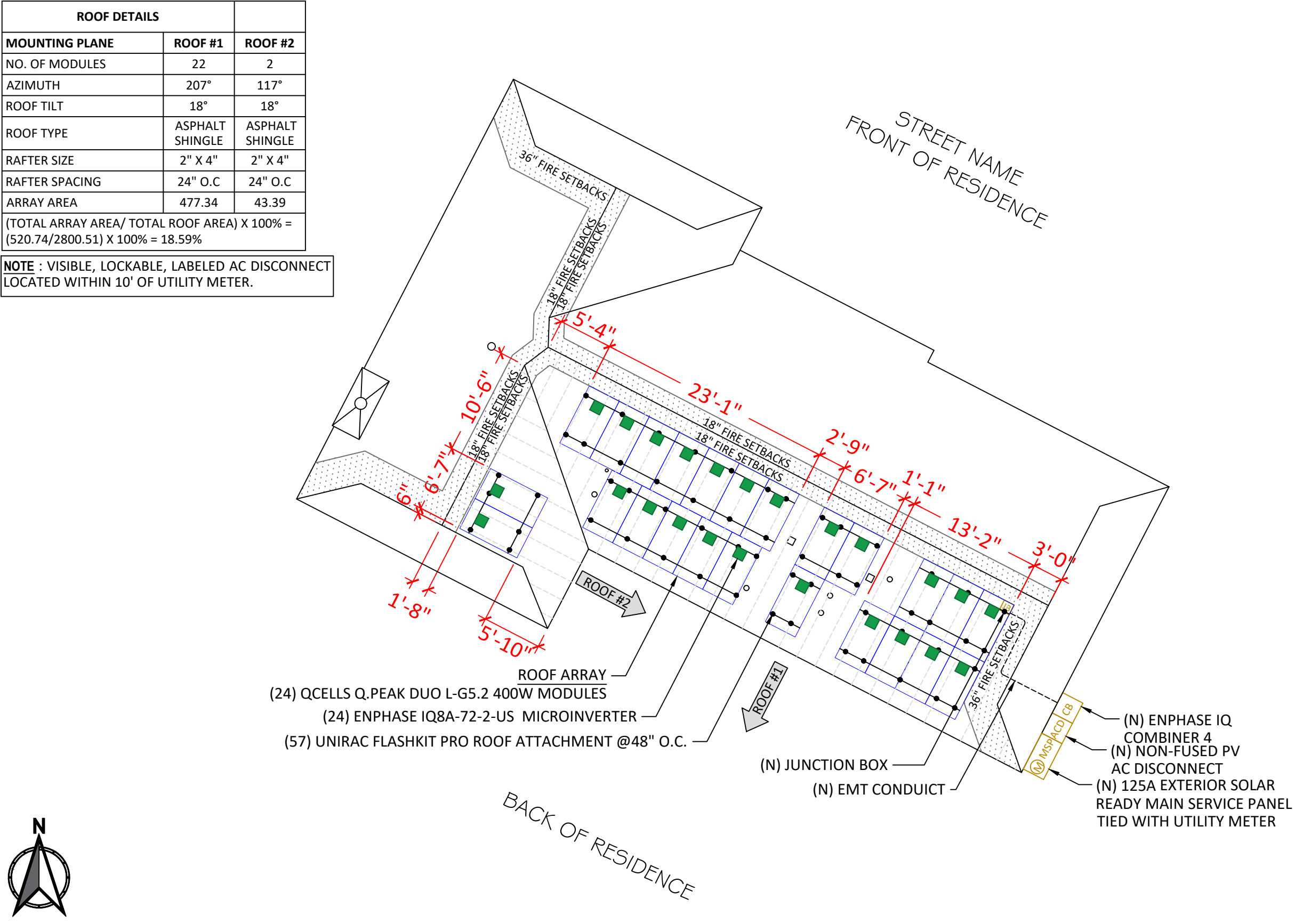


CONTRACTOR INFO	DATE	08/15/2025	CLIENT NAME RESIDENCE 1234 STREET NAME, RENO, NV 9874 APN NUMBER: 147852369	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	PV-1 SHEET NO. SITE PLAN
	PROJECT ID	SP-1						
	CREATED BY	TV						
	CHECKED BY	DS						
	SCALE	1/32" = 1'-0"						
HOMEOWNER INFO	SHEET SIZE	11"X 17"					SHEET INFORMATION	
		ANSI B						

ROOF DETAILS		
MOUNTING PLANE	ROOF #1	ROOF #2
NO. OF MODULES	22	2
AZIMUTH	207°	117°
ROOF TILT	18°	18°
ROOF TYPE	ASPHALT SHINGLE	ASPHALT SHINGLE
RAFTER SIZE	2" X 4"	2" X 4"
RAFTER SPACING	24" O.C	24" O.C
ARRAY AREA	477.34	43.39
(TOTAL ARRAY AREA/ TOTAL ROOF AREA) X 100% = (520.74/2800.51) X 100% = 18.59%		

NOTE : VISIBLE, LOCKABLE, LABELED AC DISCONNECT LOCATED WITHIN 10' OF UTILITY METER.

LEGEND	
	UTILITY METER
	MAIN SERVICE PANEL
	ENPHASE IQ COMBINER 4
	VISIBLE LOCKABLE, LABELLED PV AC DISCONNECT
	JUNCTION BOX
	IQ8A-72-2-US MICROINVERTER
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	UNIRAC FLASHKIT PRO
	UNIRAC SOLARMOUNT LT
	RAFTER
	TRENCH
	CHIMNEY



NOTES: ROOF ACCESS, PATHWAYS AND SPACING REQUIREMENTS SHALL BE PROVIDED IN ACCORDANCE WITH SECTIONS 1205.2.1 THROUGH 1205.3.3. PATHWAYS SHALL BE OVER AREAS CAPABLE OF SUPPORTING FIRE FIGHTERS ACCESSING THE ROOF. PATHWAYS SHALL BE LOCATED IN AREAS WITH MINIMAL OBSTRUCTIONS, SUCH AS VENT PIPES, CONDUIT OR MECHANICAL EQUIPMENT.

NOTES: ATTACHMENT POINTS ARE NOT EXACT. CONTRACTOR MAY NEED TO ADJUST MOUNT LOCATION. EXISTING ROOF CONDITION & RAFTER LOCATIONS ARE SUBJECT TO FIELD VERIFICATION. ACTUAL LOCATION MAY DIFFER. INSTALL PER MANUFACTURER SPECIFICATIONS. SEE SUPPLEMENTAL DOCUMENTS FOR ATTACHMENT & STRUCTURAL DOCUMENTATION.

CONTRACTOR INFO	DATE	08/15/2025
	PROJECT ID	SP-1
	CREATED BY	TV
	CHECKED BY	DS
	SCALE	1/10" = 1'-0"
SHEET SIZE	11"X 17"	
	ANSI B	

HOMEOWNER INFO	CLIENT NAME RESIDENCE	
	1234 STREET NAME, RENO, NV 9874	
	APN NUMBER: 147852369	

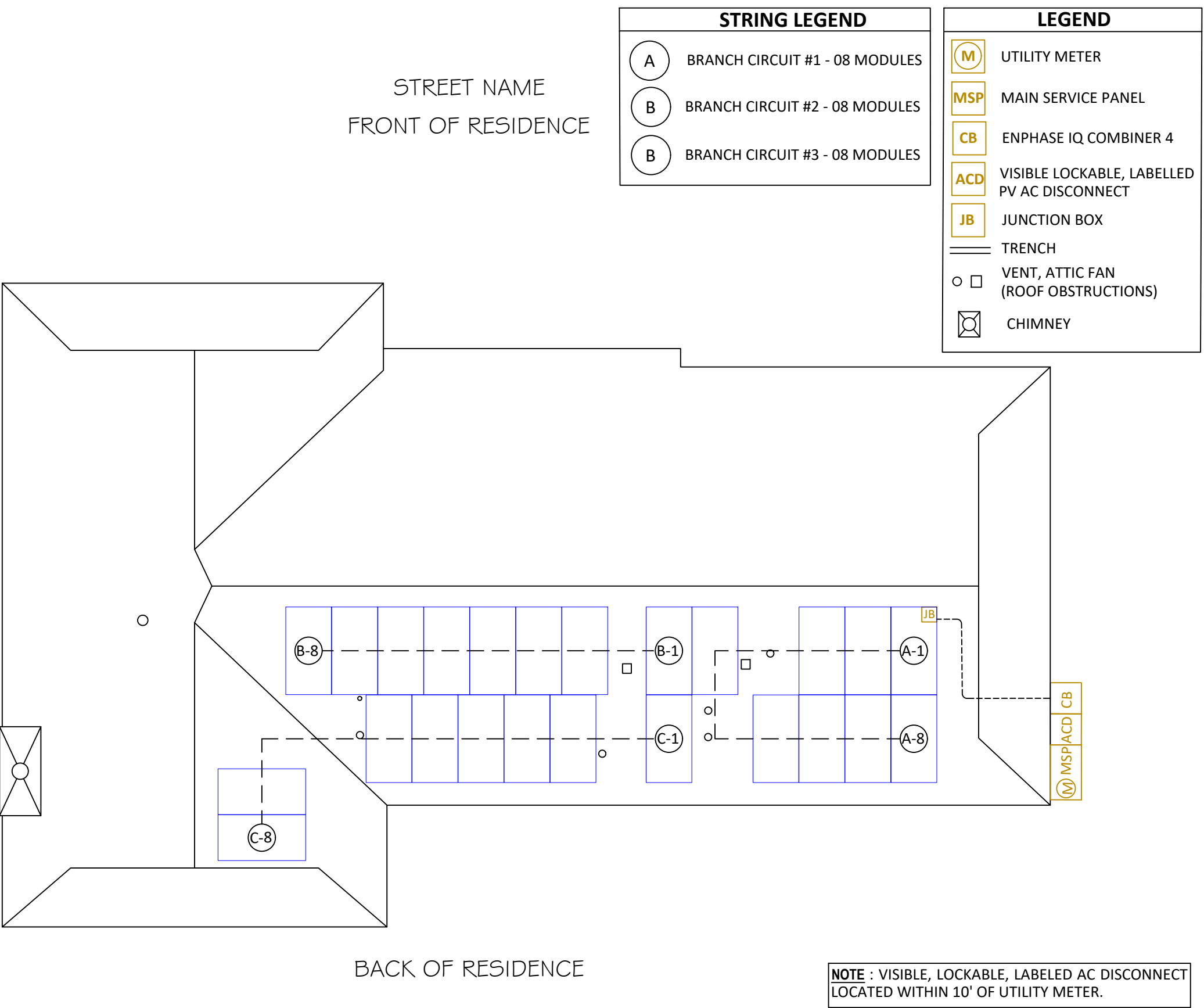
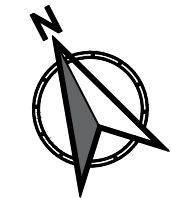
#	REVISION/UPDATE	DATE

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PV-2
SHEET NO.
ROOF ATTACHMENT DETAILS
SHEET INFORMATION

BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
SOLAR PV MODULE	24	QCELLS Q.PEAK DUO L-G5.2 400W MODULES
MICROINVERTER	24	ENPHASE IQ8A-72-2-US MICROINVERTERS
JUNCTION BOX	1	JUNCTION BOX, 600V, NEMA 3R, UL LISTED
COMBINER BOX	1	ENPHASE IQ COMBINER 4 (X-IQ-AM1-240-4)
SOLAR READY PANEL	1	EATON MBE1224PVL125S SOLAR READY PANEL, 125A RATED, 1-PH, 3-W, 240V
WATER TIGHT CAPS	11	WATER TIGHT CAPS
BRANCH TERMINATOR	3	BRANCH TERMINATOR
ATTACHMENT	57	UNIRAC FLASHKIT PRO
ATTACHMENT	57	3/8 "-16 SS T-BOLT
RAIL	12	UNIRAC SOLARMOUNT LT UL2703 (168")
BONDED SPLICE	4	BONDED SPLICE
CLAMPS	16	UNIVERSAL FASTENING OBJECTS (UFO)
CLAMPS	28	STOPPER SLEEVE
GROUNDING LUG	7	GROUNDING LUG

ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	24	
Module Weight	51.8	LBS
Total Module (Array) Weight	1243.20	LBS
Number of Attachment point	57	
Mounting System Weight	0.75	LBS
(Per Module)		
Mounting System Weight	42.75	LBS
Total System Weight	1285.95	LBS
(Module Weight + Mounting System Weight)		
Weight at Each Attachment Point	21.81	LBS
(Array Weight / Number of Attachment Point)		
Module Area (79.3" x 39.4")	21.70	SqFt
New Array Area	520.74	SqFt
Distributed Load	2.47	Per SqFt
(Total System Weight / Total Array Area)		
Total Roof Area	2800.51	SqFt
Total Percentage of Roof Covered	18.59%	
Total Array Area / Total Roof Area)*100		



STRING LEGEND	
A	BRANCH CIRCUIT #1 - 08 MODULES
B	BRANCH CIRCUIT #2 - 08 MODULES
B	BRANCH CIRCUIT #3 - 08 MODULES

LEGEND	
M	UTILITY METER
MSP	MAIN SERVICE PANEL
CB	ENPHASE IQ COMBINER 4
ACD	VISIBLE LOCKABLE, LABELLED PV AC DISCONNECT
JB	JUNCTION BOX
—	TRENCH
○ □	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
⊗	CHIMNEY

CONTRACTOR INFO	DATE	08/15/2025
	PROJECT ID	SP-1
	CREATED BY	TV
	CHECKED BY	DS
	SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"	
	ANSI B	

DATE	08/15/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	DS
SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

CLIENT NAME RESIDENCE 1234 STREET NAME, RENO, NV 9874 APN NUMBER: 147852369	HOMEOWNER INFO	

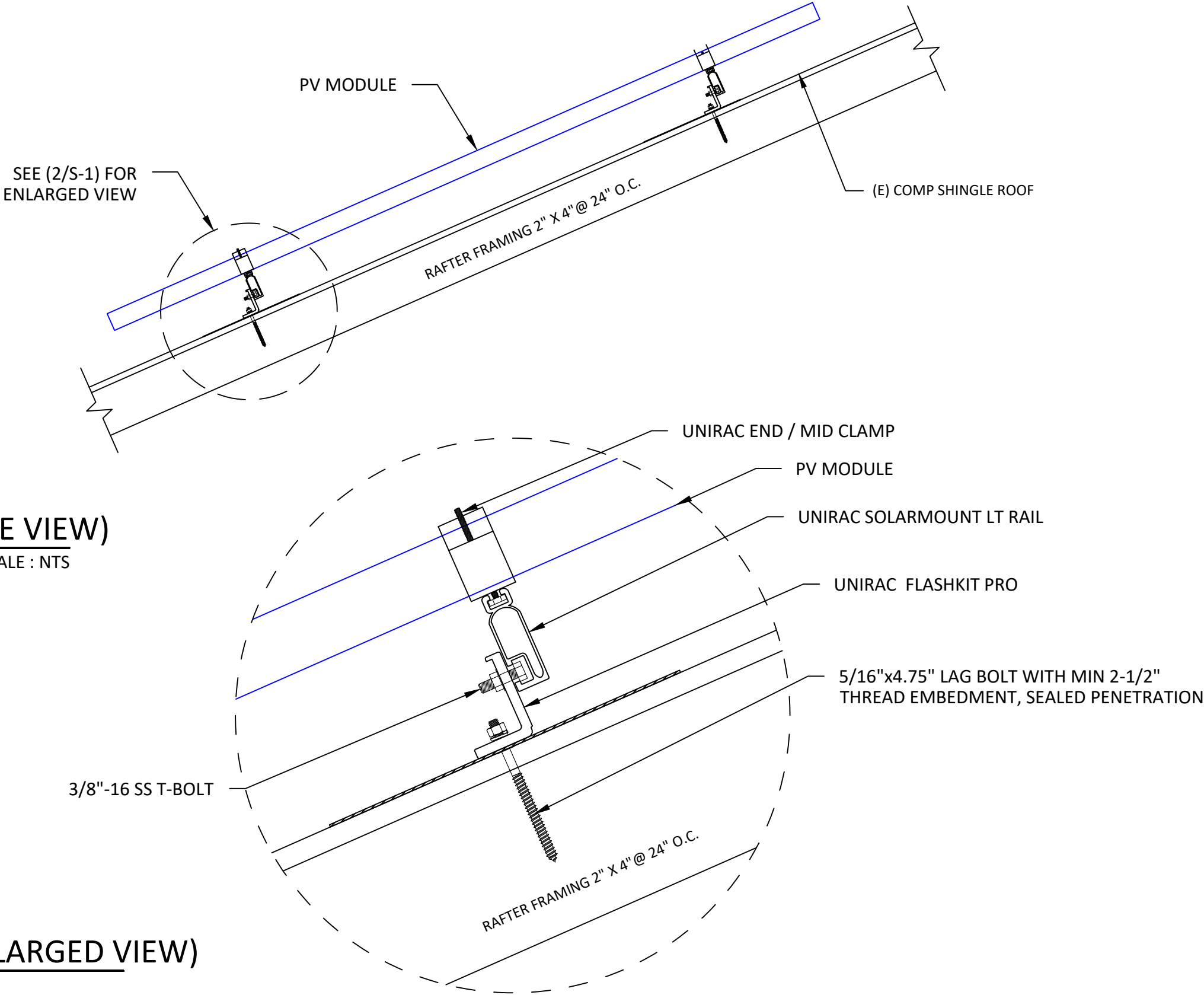
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY		

PV-3 SHEET NO. BOM & STRING LAYOUT SHEET INFORMATION		

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE UNIRAC SOLARMOUNT LT SYSTEM WITH UNIRAC FLASHKIT PRO ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. THE PROPOSED PV SYSTEM ADDS 2.47 PSF TO THE ROOF FRAMING SYSTEM.
- 4. ROOF LIVE LOAD = 20 PSF.
- 5. GROUND SNOW LOAD = 72 PSF
- 6. WIND SPEED = 96 MPH
- 7. EXPOSURE CATEGORY = B
- 8. RISK CATEGORY = II



1 ATTACHMENT DETAIL (SIDE VIEW)

SCALE : NTS

ATTACHMENT DETAIL (ENLARGED VIEW)

CONTRACTOR INFO

DATE	08/15/2025
PROJECT ID	SP-1
CREATED BY	TV
CHECKED BY	DS
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT NAME RESIDENCE
1234 STREET NAME, RENO, NV 9874
APN NUMBER: 147852369

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-1
SHEET NO.
STRUCTURAL
DETAILED
DIAGRAM
SHEET INFORMATION

PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(24) MODULES x 400W	(24) INVERTER x 349W
9.600 kW	8.376 kW
INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL	ENPHASE IQ8A-72-2-US
NOMINAL OUTPUT VOLTAGE	240VAC
CONTINUOUS OUTPUT CURRENT	1.45A

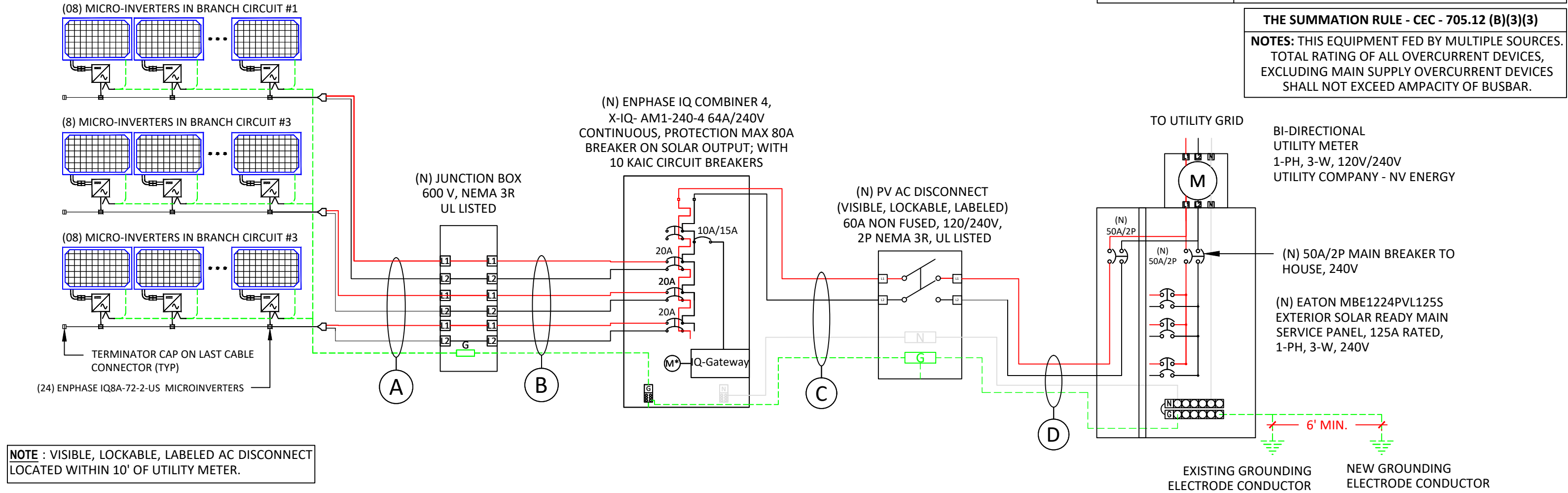
AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-15°C
AMBIENT TEMP (HIGH TEMP 2%)	35°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	96°C
CONDUCTOR TEMPERATURE RATED	57°C
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.27%/°C

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	QCELLS Q.PEAK DUO L-G5.2 400W
VMP	41.04V
IMP	9.75A
VOC	49.00V
ISC	10.24A
TEMP. COEFF. VOC	-0.27 %/°C
MODULE DIMENSION	79.3" (L) x 39.4" (W)
PANEL WATTAGE	400W

PROJECT INFORMATION	
(#) PV MODULES	(24) QCELLS Q.PEAK DUO L-G5.2 400W
(#)MICRO-INVERTER	(24) ENPHASE IQ8A-72-2-US
(#) IQ COMBINER	(1) ENPHASE IQ COMBINER 4
SYSTEM OUTPUT CURRENT	34.80A
(#) BRANCH CIRCUITS	3 BRANCH CIRCUITS OF 8 MODULES
POINT OF INTERCONNECTION	50A LOAD SIDE BREAKER IN EXTERIOR NEW SOLAR READY MAIN SERVICE PANEL WITH 125A BUSBAR, 120/240V, 3W, 1φ
UTILITY	NV ENERGY
AHJ	WASHOE COUNTY

THE SUMMATION RULE - CEC - 705.12 (B)(3)(3)

NOTES: THIS EQUIPMENT FED BY MULTIPLE SOURCES. TOTAL RATING OF ALL OVERCURRENT DEVICES, EXCLUDING MAIN SUPPLY OVERCURRENT DEVICES SHALL NOT EXCEED AMPACITY OF BUSBAR.



ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (INVERTER CONTINUOUS CURRENT x 1.25)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	QCABLE		-	1	2	N/A	N/A	6 AWG	BARE COPPER	0.71	(57°C)	N/A	11.6A	x	1.25	=	14.5A	N/A	N/A	33FT	0.36%
B	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	3	6	29.59 %	20A	8 AWG	THWN-2, COPPER	0.94	(35°C)	0.8	11.6A	x	1.25	=	14.5A	35A	35A x 0.8 x 0.88 = 24.6A	25FT	0.31%
C	8 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	1	3	26.73 %	N/A	8 AWG	THWN-2, COPPER	0.94	(35°C)	1	34.8A	x	1.25	=	43.5A	50A	50A x 1 x 0.88 = 44.0A	5FT	0.12%
D	8 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	1	3	26.73%	50A	8 AWG	THWN-2, COPPER	0.94	(35°C)	1	34.8A	x	1.25	=	43.5A	50A	50A x 1 x 0.88 = 44.0A	5FT	0.13%

CONTRACTOR INFO	DATE	08/15/2025	CLIENT NAME RESIDENCE 1234 STREET NAME, RENO, NV 9874 APN NUMBER: 147852369	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-1 SHEET NO. ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION
	PROJECT ID	SP-1						
	CREATED BY	TV						
	CHECKED BY	DS						
	SCALE	NTS						
HOMEOWNER INFO	SHEET SIZE	11"X 17"						
		ANSI B						

1

!WARNING

ELECTRIC SHOCK HAZARD

DO NOT TOUCH TERMINALS.
TERMINALS ON BOTH LINE AND LOAD SIDES MAY
BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT/INVERTER/
AC COMBINER
2023 NEC 690.13(B)

2

!WARNING

ELECTRIC SHOCK HAZARD

IF GROUND FAULT IS INDICATED
ALL NORMALLY GROUNDED
CONDUCTORS MAY BE
UNGROUND AND ENERGIZED

LABEL LOCATION:
INVERTER(S), ENPHASE ENVOY ENCLOSURE (IF
APPLICABLE).
PER CODE(S): NEC 2023: 690.5(C)

3

PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT34.80A

NOMINAL OPERATING AC VOLTAGE240 VAC

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT
2023 NEC 690.53

4

RAPID SHUTDOWN SWITCH
FOR SOLAR PV SYSTEM

LABEL LOCATION:
INVERTER
AT OR WITHIN 3' OF THE DC COMBINER
SWITCH
2023 NEC 690.56(C)(3)

5

SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM
AND REDUCE
SHOCK HAZARD
IN THE ARRAY

SOLAR ELECTRIC
PV PANELS

LABEL LOCATION:
MAIN SERVICE PANEL
IF MSD IS OUTSIDE PLACE IT THERE / IF
MSD IS INSIDE PLACE ON THE AC DISCONNECT
2023 NEC 690.56(C)

6

!WARNING

PHOTOVOLTAIC SYSTEM
COMBINER PANEL

DO NOT ADD LOADS

LABEL LOCATION:
PHOTOVOLTAIC AC COMBINER (IF APPLICABLE).
PER CODE(S): NEC 2023: 705.12(B)(3)(2)

7

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES.
TOTAL RATING OF ALL OVERCURRENT DEVICES,
EXCLUDING MAIN SUPPLY OVERCURRENT DEVICES
SHALL NOT EXCEED AMPACITY OF BUSBAR.

LABEL LOCATION:
MAIN SERVICE PANEL
REQ'D BY: NEC 705.12(B)(3)(3)

8

PHOTOVOLTAIC SYSTEM
UTILITY DISCONNECT SWITCH

LABEL LOCATION:
AC DISCONNECT
2023 NEC 690.54

9

!CAUTION

DUAL POWER SOURCE
SECOND SOURCE IS
PHOTOVOLTAIC

LABEL LOCATION:
SUB PANEL MAIN SERVICE PANEL,
METER/MAIN AND PV AC DISCONNECT
2023 NEC 705.10

CAUTION!

POWER TO THIS BUILDING IS ALSO SUPPLIED FROM THE
FOLLOWING SOURCES WITH DISCONNECTS LOCATED AS SHOWN.

PV ROOF ARRAY

METER-MAIN SERVICE PANEL

COMBINER BOX
PV AC DISCONNECT

STREET NAME

LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

ADHESIVE FASTENED SIGNS:

- LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
- LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
- THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
- AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.

CONTRACTOR INFO	DATE	08/15/2025	CLIENT NAME RESIDENCE 1234 STREET NAME, RENO, NV 9874 APN NUMBER: 147852369	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-2 SHEET NO. PLACARDS SHEET INFORMATION											
	PROJECT ID	SP-1		CREATED BY	TV	CHECKED BY			DS	SCALE	NTS	SHEET SIZE	11"X 17"		ANSI B	HOMEOWNER INFO			
	CREATED BY	TV		CHECKED BY	DS	SCALE			NTS	SHEET SIZE	11"X 17"		ANSI B	HOMEOWNER INFO					
	CHECKED BY	DS		SCALE	NTS	SHEET SIZE			11"X 17"		ANSI B	HOMEOWNER INFO							
	SCALE	NTS		SHEET SIZE	11"X 17"				ANSI B	HOMEOWNER INFO									
	SHEET SIZE	11"X 17"			ANSI B	HOMEOWNER INFO													
		ANSI B																	
HOMEOWNER INFO																			



Q.PEAK DUO L-G5.2

380-405

ENDURING HIGH PERFORMANCE



Q.ANTUM TECHNOLOGY: LOW LEVELISED COST OF ELECTRICITY

Higher yield per surface area, lower BOS costs, higher power classes, and an efficiency rate of up to 20.3%.



INNOVATIVE ALL-WEATHER TECHNOLOGY

Optimal yields, whatever the weather with excellent low-light and temperature behaviour.



ENDURING HIGH PERFORMANCE

Long-term yield security with Anti LID Technology, Anti PID Technology¹, Hot-Spot Protect and Traceable Quality Tra.Q™.



EXTREME WEATHER RATING

High-tech aluminium alloy frame, certified for high snow (5400 Pa) and wind loads (2400 Pa).



A RELIABLE INVESTMENT

Inclusive 12-year product warranty and 25-year linear performance warranty².



STATE OF THE ART MODULE TECHNOLOGY

Q.ANTUM DUO combines cutting edge cell separation and innovative wiring with Q.ANTUM Technology.

¹ APT test conditions according to IEC/TS 62804-1:2015, method B (-1500 V, 168 h)

² See data sheet on rear for further information.

THE IDEAL SOLUTION FOR:



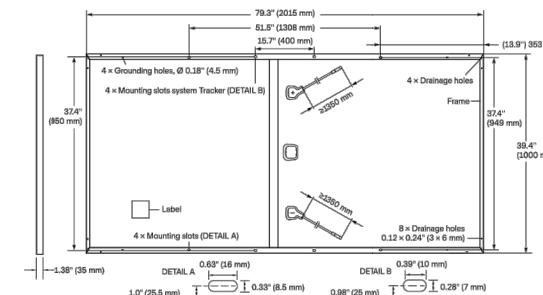
Rooftop arrays on commercial/industrial buildings



Ground-mounted solar power plants

MECHANICAL SPECIFICATION

Format	79.3 in x 39.4 in x 1.38 in (including frame) (2015 mm x 1000 mm x 35 mm)
Weight	51.8 lbs (23.5 kg)
Front Cover	0.13 in (3.2 mm) thermally pre-stressed glass with anti-reflection technology
Back Cover	Composite film
Frame	Anodized aluminum
Cell	6 x 24 monocrystalline Q.ANTUM solar half cells
Junction Box	2.09-3.98 x 1.26-2.36 x 0.59-0.71 in (53-101 x 32-60 x 15-18 mm), Protection class IP67, with bypass diodes
Cable	4 mm ² Solar cable; (+) ≥53.1 in (1350 mm), (-) ≥53.1 in (1350 mm)
Connector	Stäubli MC4-Evo2, Hanwha Q CELLS HQC4, Amphenol UTX, Renhe 05-8, JMTHY JM601A; Tongling Cable01S-F, IP68 or Friends PV2e; IP67

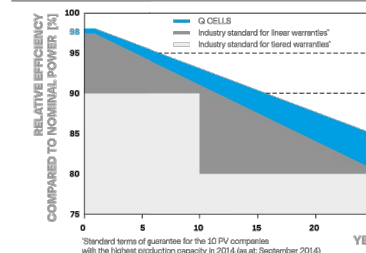


ELECTRICAL CHARACTERISTICS

POWER CLASS		380	385	390	395	400	405
MINIMUM PERFORMANCE AT STANDARD TEST CONDITIONS, STC ¹ (POWER TOLERANCE +5 W / -0 W)							
Minimum	Power at MPP ¹	P _{MPP} [W]	380	385	390	395	405
	Short Circuit Current ¹	I _{SC} [A]	10.05	10.10	10.14	10.19	10.28
	Open Circuit Voltage ¹	V _{OC} [V]	47.95	48.21	48.48	48.74	49.00
	Current at MPP	I _{MPP} [A]	9.57	9.61	9.66	9.70	9.79
	Voltage at MPP	V _{MPP} [V]	39.71	40.05	40.38	40.71	41.04
	Efficiency ¹	η [%]	≥18.9	≥19.1	≥19.4	≥19.6	≥19.9
MINIMUM PERFORMANCE AT NORMAL OPERATING CONDITIONS, NMOT ²							
Minimum	Power at MPP	P _{MPP} [W]	284.4	288.2	291.9	295.6	299.4
	Short Circuit Current	I _{SC} [A]	8.10	8.14	8.17	8.21	8.25
	Open Circuit Voltage	V _{OC} [V]	45.21	45.46	45.71	45.96	46.21
	Current at MPP	I _{MPP} [A]	7.53	7.57	7.60	7.64	7.67
	Voltage at MPP	V _{MPP} [V]	37.77	38.08	38.40	38.71	39.02

¹ Measurement tolerances P_{MPP} ±3%; I_{SC}, V_{OC} ±5% at STC: 1000 W/m², 25 ±2 °C, AM 1.5 according to IEC 60904-3 • ² 800 W/m², NMOT, spectrum AM 1.5

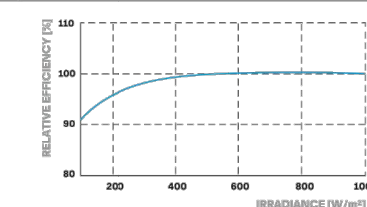
Q CELLS PERFORMANCE WARRANTY



At least 98% of nominal power during first year. Thereafter max. 0.54% degradation per year. At least 93.1% of nominal power up to 10 years. At least 85% of nominal power up to 25 years.

All data within measurement tolerances. Full warranties in accordance with the warranty terms of the Q CELLS sales organisation of your respective country.

PERFORMANCE AT LOW IRRADIANCE



Typical module performance under low irradiance conditions in comparison to STC conditions (25 °C, 1000 W/m²)

TEMPERATURE COEFFICIENTS

Temperature Coefficient of I _{SC}	α	[%/K]	+0.04	Temperature Coefficient of V _{OC}	β	[%/K]	-0.27
Temperature Coefficient of P _{MPP}	γ	[%/K]	-0.36	Nominal Module Operating Temperature	NMOT	[°F]	109 ± 5.4 (43 ± 3 °C)

PROPERTIES FOR SYSTEM DESIGN

Maximum System Voltage V _{sys}	[V]	1500 (IEC) / 1500 (UL)	PV module classification	Class II
Maximum Series Fuse Rating	[A DC]	20	Fire Rating based on ANSI / UL 61730	TYPE 1
Max. Design Load, Push / Pull ³	[lbs / ft ²]	75 (3600 Pa) / 33 (1600 Pa)	Permitted Module Temperature on Continuous Duty	-40 °F up to +185 °F (-40 °C up to +85 °C)
Max. Test Load, Push / Pull ³	[lbs / ft ²]	113 (5400 Pa) / 50 (2400 Pa)		

³ See Installation Manual

QUALIFICATIONS AND CERTIFICATES

UL 61730, CE-compliant,
IEC 61215:2016,
IEC 61730:2016,
U.S. Patent No. 9,893,215
(solar cells)



PACKAGING INFORMATION

	81.1 in 2060 mm	40.9 in 1040 mm	47.1 in 1196 mm	1588 lbs 721 kg	24 pallets	22 pallets	29 modules
Horizontal packaging							
Vertical packaging	84.6 in 2090 mm	45.3 in 1150 mm	46.1 in 1170 mm	1635 lbs 742 kg	27 pallets	22 pallets	29 modules

Note: Installation instructions must be followed. See the installation and operating manual or contact our technical service department for further information on approved installation and use of this product. Q CELLS supplies solar modules in two different stacking methods, depending on the location of manufacture (modules are packed horizontally or vertically). You can find more detailed information in the document "Packaging and Transport Information", available from Q CELLS.

Hanwha Q CELLS America Inc.

400 Spectrum Center Drive, Suite 1400, Irvine, CA 92618, USA | TEL +1 949 748 59 96 | EMAIL inquiry@us.q-cells.com | WEB www.q-cells.us

Engineered in Germany





IQ8M and IQ8A Microinverters

Our newest IQ8 Microinverters are the industry’s first microgrid-forming, software defined microinverters with split-phase power conversion capability to convert DC power to AC power efficiently. The brain of the semiconductor-based microinverter is our proprietary application specific integrated circuit (ASIC) which enables the microinverter to operate in grid-tied or off-grid modes. This chip is built in advanced 55nm technology with high speed digital logic and has superfast response times to changing loads and grid events, alleviating constraints on battery sizing for home energy systems.



Part of the Enphase Energy System, IQ8 Series Microinverters integrate with the IQ Battery, IQ Gateway, and the Enphase App monitoring and analysis software.



Connect PV modules quickly and easily to IQ8 Series Microinverters using the included Q-DCC-2 adapter cable with plug-n-play MC4 connectors.



IQ8 Series Microinverters redefine reliability standards with more than one million cumulative hours of power-on testing, enabling an industry-leading limited warranty of up to 25 years.



IQ8 Series Microinverters are UL listed as PV Rapid Shutdown Equipment and conform with various regulations, when installed according to manufacturer’s instructions.

*Only when installed with IQ System Controller 2, meets UL 1741.
**IQ8M and IQ8A support split-phase, 240V installations only.

Easy to install

- Lightweight and compact with plug-n-play connectors
- Power Line Communication (PLC) between components
- Faster installation with simple two-wire cabling

High productivity and reliability

- Produce power even when the grid is down*
- More than one million cumulative hours of testing
- Class II double-insulated enclosure
- Optimized for the latest high-powered PV modules

Microgrid-forming

- Complies with the latest advanced grid support**
- Remote automatic updates for the latest grid requirements
- Configurable to support a wide range of grid profiles
- Meets CA Rule 21 (UL 1741-SA) and IEEE 1547:2018 (UL 1741-SB 3rd Ed.)

Note:

IQ8 Microinverters cannot be mixed together with previous generations of Enphase microinverters (IQ7 Series, IQ6 Series, etc) in the same system.

IQ8M and IQ8A Microinverters

INPUT DATA (DC)		I08M-72-2-US	I08A-72-2-US
Commonly used module pairings¹	W	260 – 460	295 – 500
Module compatibility		54-cell / 108 half-cell, 60-cell / 120 half-cell, 66-cell / 132 half-cell and 72-cell / 144 half-cell	
MPPT voltage range	V	30 – 45	32 – 45
Operating range	V	16 – 58	
Min. / Max. start voltage	V	22 / 58	
Max. input DC voltage	V	60	
Max. continuous input DC current	A	12	
Max. input DC short-circuit current	A	25	
Max. module I _{sc}	A	20	
Overvoltage class DC port		II	
DC port backfeed current	mA	0	
PV array configuration		1 x 1 Ungrounded array; No additional DC side protection required; AC side protection requires max 20A per branch circuit	
OUTPUT DATA (AC)		I08M-72-2-US	I08A-72-2-US
Peak output power	VA	330	366
Max. continuous output power	VA	325	349
Nominal (L-L) voltage / range²	V	240 / 211 – 264	
Max. continuous output current	A	1.35	1.45
Nominal frequency	Hz	60	
Extended frequency range	Hz	47 – 68	
AC short circuit fault current over 3 cycles	Arms	2	
Max. units per 20 A (L-L) branch circuit³		11	
Total harmonic distortion		<5%	
Overvoltage class AC port		III	
AC port backfeed current	mA	30	
Power factor setting		1.0	
Grid-tied power factor (adjustable)		0.85 leading – 0.85 lagging	
Peak efficiency	%	97.8	97.7
CEC weighted efficiency	%	97.5	97
Night-time power consumption	mW	60	
MECHANICAL DATA			
Ambient temperature range		-40°C to +60°C (-40°F to +140°F)	
Relative humidity range		4% to 100% (condensing)	
DC Connector type		MC4	
Dimensions (H x W x D)		212 mm (8.3”) x 175 mm (6.9”) x 30.2 mm (1.2”)	
Weight		1.08 kg (2.38 lbs)	
Cooling		Natural convection – no fans	
Approved for wet locations		Yes	
Pollution degree		PD3	
Enclosure		Class II double-insulated, corrosion resistant polymeric enclosure	
Environ. category / UV exposure rating		NEMA Type 6 / outdoor	
COMPLIANCE			
Certifications	CA Rule 21 (UL 1741-SA), UL 62109-1, IEEE 1547:2018 (UL 1741-SB 3 rd Ed.), FCC Part 15 Class B, ICES-0003 Class B, CAN / CSA-C22.2 NO. 107.1-01 This product is UL Listed as PV Rapid Shutdown Equipment and conforms with NEC 2014, NEC 2017, and NEC 2020 section 690.12 and C22.1-2018 Rule 64-218 Rapid Shutdown of PV Systems, for AC and DC conductors, when installed according to manufacturer's instructions.		

(1) Pairing PV modules with wattage above the limit may result in additional clipping losses. See the compatibility calculator at <https://link.enphase.com/module-compatibility>. (2) Nominal voltage range can be extended beyond nominal if required by the utility. (3) Limits may vary. Refer to local requirements to define the number of microinverters per branch in your area.

Enphase
IQ Combiner 4/4C
X-IQ-AM1-240-4
X-IQ-AM1-240-4C



The **Enphase IQ Combiner 4/4C** with Enphase IQ Gateway and integrated LTE-M1 cell modem (included only with IQ Combiner 4C) consolidates interconnection equipment into a single enclosure and streamlines IQ microinverters and storage installations by providing a consistent, pre-wired solution for residential applications. It offers up to four 2-pole input circuits and Eaton BR series busbar assembly.

Smart

- Includes IQ Gateway for communication and control
- Includes Enphase Mobile Connect cellular modem (CELLMODEM-M1-06-SP-05), included only with IQ Combiner 4C
- Includes solar shield to match Enphase IQ Battery aesthetics and deflect heat
- Flexible networking supports Wi-Fi, Ethernet, or cellular
- Optional AC receptacle available for PLC bridge
- Provides production metering and consumption monitoring

Simple

- Centered mounting brackets support single stud mounting
- Supports bottom, back and side conduit entry
- Up to four 2-pole branch circuits for 240 VAC plug-in breakers (not included)
- 80A total PV or storage branch circuits

Reliable

- Durable NRTL-certified NEMA type 3R enclosure
- Five-year limited warranty
- Two years labor reimbursement program coverage included for both the IQ Combiner SKU's
- UL listed



To learn more about Enphase offerings, visit enphase.com

Enphase IQ Combiner 4/4C

MODEL NUMBER	
IQ Combiner 4 (X-IQ-AM1-240-4)	IQ Combiner 4 with Enphase IQ Gateway printed circuit board for integrated revenue grade PV production metering (ANSI C12.20 +/- 0.5%) and consumption monitoring (+/- 2.5%). Includes a silver solar shield to match the IQ Battery system and IQ System Controller 2 and to deflect heat.
IQ Combiner 4C (X-IQ-AM1-240-4C)	IQ Combiner 4C with Enphase IQ Gateway printed circuit board for integrated revenue grade PV production metering (ANSI C12.20 +/- 0.5%) and consumption monitoring (+/- 2.5%). Includes Enphase Mobile Connect cellular modem (CELLMODEM-M1-06-SP-05), a plug-and-play industrial-grade cell modem for systems up to 60 microinverters. (Available in the US, Canada, Mexico, Puerto Rico, and the US Virgin Islands, where there is adequate cellular service in the installation area.) Includes a silver solar shield to match the IQ Battery and IQ System Controller and to deflect heat.
ACCESSORIES AND REPLACEMENT PARTS (not included, order separately)	
Ensemble Communications Kit COMMS-CELLMODEM-M1-06 CELLMODEM-M1-06-SP-05 CELLMODEM-M1-06-AT-05	- Includes COMMS-KIT-01 and CELLMODEM-M1-06-SP-05 with 5-year Sprint data plan for Ensemble sites - 4G based LTE-M1 cellular modem with 5-year Sprint data plan - 4G based LTE-M1 cellular modem with 5-year AT&T data plan
Circuit Breakers BRK-10A-2-240V BRK-15A-2-240V BRK-20A-2P-240V BRK-15A-2P-240V-B BRK-20A-2P-240V-B	Supports Eaton BR210, BR215, BR220, BR230, BR240, BR250, and BR260 circuit breakers. Circuit breaker, 2 pole, 10A, Eaton BR210 Circuit breaker, 2 pole, 15A, Eaton BR215 Circuit breaker, 2 pole, 20A, Eaton BR220 Circuit breaker, 2 pole, 15A, Eaton BR215B with hold down kit support Circuit breaker, 2 pole, 20A, Eaton BR220B with hold down kit support
EPLC-01	Power line carrier (communication bridge pair), quantity - one pair
XA-SOLARSHIELD-ES	Replacement solar shield for IQ Combiner 4/4C
XA-PLUG-120-3	Accessory receptacle for Power Line Carrier in IQ Combiner 4/4C (required for EPLC-01)
XA-ENV-PCBA-3	Replacement IQ Gateway printed circuit board (PCB) for Combiner 4/4C
X-IQ-NA-HD-125A	Hold down kit for Eaton circuit breaker with screws.
ELECTRICAL SPECIFICATIONS	
Rating	Continuous duty
System voltage	120/240 VAC, 60 Hz
Eaton BR series busbar rating	125 A
Max. continuous current rating	65 A
Max. continuous current rating (input from PV/storage)	64 A
Max. fuse/circuit rating (output)	90 A
Branch circuits (solar and/or storage)	Up to four 2-pole Eaton BR series Distributed Generation (DG) breakers only (not included)
Max. total branch circuit breaker rating (input)	80A of distributed generation / 95A with IQ Gateway breaker included
Envoy breaker	10A or 15A rating GE/Siemens/Eaton included
Production metering CT	200 A solid core pre-installed and wired to IQ Gateway
Consumption monitoring CT (CT-200-SPLIT)	A pair of 200 A split core current transformers
MECHANICAL DATA	
Dimensions (WxHxD)	37.5 x 49.5 x 16.8 cm (14.75" x 19.5" x 6.63"). Height is 21.06" (53.5 cm) with mounting brackets.
Weight	7.5 kg (16.5 lbs)
Ambient temperature range	-40° C to +46° C (-40° to 115° F)
Cooling	Natural convection, plus heat shield
Enclosure environmental rating	Outdoor, NRTL-certified, NEMA type 3R, polycarbonate construction
Wire sizes	• 20 A to 50 A breaker inputs: 14 to 4 AWG copper conductors • 60 A breaker branch input: 4 to 1/0 AWG copper conductors • Main lug combined output: 10 to 2/0 AWG copper conductors • Neutral and ground: 14 to 1/0 copper conductors Always follow local code requirements for conductor sizing.
Altitude	To 2000 meters (6,560 feet)
INTERNET CONNECTION OPTIONS	
Integrated Wi-Fi	802.11b/g/n
Cellular	CELLMODEM-M1-06-SP-05, CELLMODEM-M1-06-AT-05 (4G based LTE-M1 cellular modem). Note that an Enphase Mobile Connect cellular modem is required for all Ensemble installations.
Ethernet	Optional, 802.3, Cat5E (or Cat 6) UTP Ethernet cable (not included)
COMPLIANCE	
Compliance, IQ Combiner	UL 1741, CAN/CSA C22.2 No. 107.1, 47 CFR, Part 15, Class B, ICES 003 Production metering: ANSI C12.20 accuracy class 0.5 (PV production) Consumption metering: accuracy class 2.5
Compliance, IQ Gateway	UL 60601-1/CANCSA 22.2 No. 61010-1

To learn more about Enphase offerings, visit enphase.com

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Maximize solar potential, while minimizing product cost

With the new Line Side Solar Meter Breaker, the solar connection is made before the distribution, allowing for maximum solar rating without the need to oversize the distribution bus. This helps minimize the cost of the unit and also eliminates the need to upsize the utility service to match the maximum distribution bus rating.

Cost Effective

Whether for new installation or renovation, the line side solar meter breaker allows for up to 125A solar without the need to up-size incoming utility wire to match the max amperage of the distribution bus.

Markets and applications

- EUSERC Meter Breaker
- Residential

Features and benefits

- Solar rating is no longer constrained by distribution bus rating (NEC690.64)
- Utility cable can be rated to match the amperage of the main breaker instead of the max bus rating
- Factory labeled for solar ready installation



Line Side Solar Meter Breaker product selection

Amperes	Main Breaker	Spaces/circuits	Additional features	Catalog number
100	Yes	12/24	Flush style	MBE1224PVL100F
100	Yes	12/24	Surface style	MBE1224PVL100S
125	Yes	12/24	Flush style	MBE1224PVL125F
125	Yes	12/24	Surface style	MBE1224PVL125S



Eaton
1000 Eaton Boulevard
Cleveland, OH 44122
United States
Eaton.com

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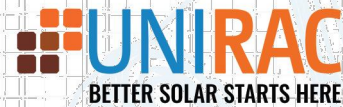
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FLASHKIT PRO FLASHKIT SWITCHBLADE



INSTALL WITH PEACE OF MIND

- Features **SHED & SEAL** technology
- Pre-kitted in packs of 10 for maximum convenience
- All the hardware in one box makes for a fast and efficient install.
- Use FLASHKIT PRO SB for less prying of shingles and less pulling of nails on the roof.

HIGH-SPEED INSTALLATION

- Faster installation in 3 simple steps, (Check the installation guide, visit UNIRAC'S website: www.unirac.com).

WATER RESISTANT

- TAS 100(A)-95 Wind Driven Rain performed by Intertek

WHY FLASHKIT PRO & FLASHKIT SWITCHBLADE?

FLASHKIT PRO and FLASHKIT PRO SWITCHBLADE (SB) offer complete attachment solutions for composition shingle roofs. Featuring Unirac's patented SHED & SEAL technology, a weatherproof system which provides the ultimate protection against roof leaks. This trusted water seal technology for composition shingle roofs is now available in standard width and in a more compact option.

KITTED WITH

- TEN (10) Flashings
- TEN (10) Continuous Slot L-Feet
- TEN (10) Hardware

FLASHKIT SB
Available in Dark
and Mill

FLASHKIT SWITCHBLADE

UNIRAC
25
YEAR
FULL-SYSTEM
WARRANTY

FLASHKIT PRO

FLASHKIT PRO
Available in Dark
and Mill

FLASHKIT PRO

INSTALLATION GUIDE



FLASHKIT PRO IS THE COMPLETE FLASHING AND ATTACHMENT SOLUTION FOR COMPOSITION ROOFS.



INSTALL FLASHKIT PRO FLASHING



INSTALL L-FOOT



ATTACH L-FOOT TO RAIL

PRE-INSTALL

- Locate roof rafters and snap chalk lines to mark the installation point for each roof attachment.
- Drill a 7/32" pilot hole at each roof attachment. Fill each pilot hole with sealant.

STEP 1 INSTALL FLASHKIT PRO FLASHING

- Add a U-shaped bead of roof sealant to the underside of the flashing with the open side of the U pointing down the roof slope. Slide the aluminum flashing underneath the row of shingles directly up slope from the pilot hole as shown. Align the indicator marks on the lower end of the flashing with the chalk lines on the roof to center the raised hole in the flashing over the pilot hole in the roof. When installed correctly, the flashing will extend under the two courses of shingles above the pilot hole.

STEP 2 INSTALL L-FOOT

- Fasten L-foot and Flashing into place by passing the included lag bolt and pre-installed stainless steel-backed EPDM washer through the L-foot EPDM grommet, and the raised hole in the flashing, into the pilot hole in the roof rafter.

- Drive the lag bolt down until the L-foot is held firmly in place. It is normal for the EPDM on the underside of the stainless steel backed EPDM washer to compress and expand beyond the outside edge of the steel washer when the proper torque is applied.

TIP:

- Use caution to avoid over-torquing the lag bolt if using an impact driver.
- Repeat Steps 1 and 2 at each roof attachment point.

STEP 3 ATTACH L-FOOT TO RAIL

- Insert the included 3/8"-16 T-bolts into the lower slot on the Rail (sold separately), spacing the bolts to match the spacing between the roof attachments.
- Position the Rail against the L-Foot and insert the threaded end of the T-Bolt through the continuous slot in the L-Foot. Apply anti-seize to bolt threads to prevent galling of the T-bolt and included 3/8" serrated flange nut. Place the 3/8" flange nut on the T-bolt and finger tighten. Repeat STEP 3 until all L-Feet are secured to the Rail with a T-bolt. Adjust the level and height of the Rail and torque each bolt to 30ft-lbs.

FASTER INSTALLATION. 25-YEAR WARRANTY.

FOR QUESTIONS OR CUSTOMER SERVICE VISIT UNIRAC.COM OR CALL (505) 248-2702

FOR QUESTIONS OR CUSTOMER SERVICE CONTACT:
505-242-6411 | SALES@UNIRAC.COM | WWW.UNIRAC.COM
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CONFORMS TO
UL2703

ISO
9001:2015
14001:2015
CERTIFIED

SOLARMOUNT



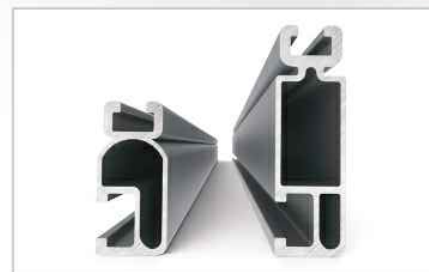
SOLARMOUNT defined the standard in solar racking. Features are designed to get installers off the roof faster. Our grounding & bonding process eliminates copper wire and grounding straps to reduce costs. Systems can be configured with standard or light rail to meet your design requirements at the lowest cost possible. The superior aesthetics package provides a streamlined clean edge for enhanced curb appeal, with no special brackets required for installation.



Now Featuring:
THE NEW FACE OF SOLAR RACKING
Superior Aesthetics Package



LOSE ALL OF THE COPPER & LUGS
System grounding through Enphase microinverters and trunk cables



SMALL IS THE NEXT NEW BIG THING
Light Rail is Fully Compatible with all SM Components



ENHANCED DESIGN & LAYOUT TOOLS
Featuring Google Map Capabilities within U-Builder

FAST INSTALLATION. SUPERIOR AESTHETICS

OPTIMIZED COMPONENTS • VERSATILITY • DESIGN TOOLS • QUALITY PROVIDER

SOLARMOUNT



OPTIMIZED COMPONENTS

INTEGRATED BONDING & PRE-ASSEMBLED PARTS

Components are pre-assembled and optimized to reduce installation steps and save labor time. Our new grounding & bonding process eliminates copper wire and grounding straps or bonding jumpers to reduce costs. Utilize the microinverter mount with a wire management clip for an easier installation.

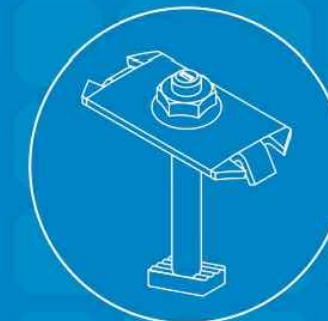
VERSATILITY

ONE PRODUCT - MANY APPLICATIONS

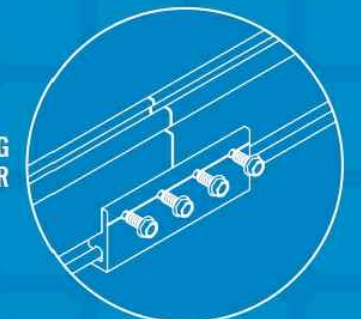
Quickly set modules flush to the roof or at a desired tilt angle. Change module orientation to portrait or landscape while securing a large variety of framed modules on flat, low slope or steep pitched roofs. Available in mill, clear and dark anodized finishes to outperform your projects financial and aesthetic aspirations.

AUTOMATED DESIGN TOOL DESIGN PLATFORM AT YOUR SERVICE

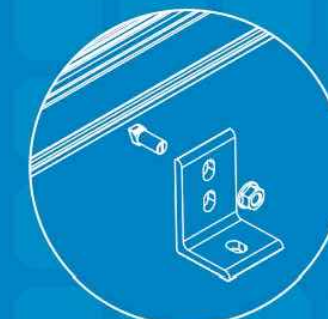
Creating a bill of materials is just a few clicks away with U-Builder, a powerful online tool that streamlines the process of designing a code compliant solar mounting system. Save time by creating a user profile, and recall preferences and projects automatically when you log in. You will enjoy the ability to share projects with customers; there's no need to print results and send to a distributor, just click and share.



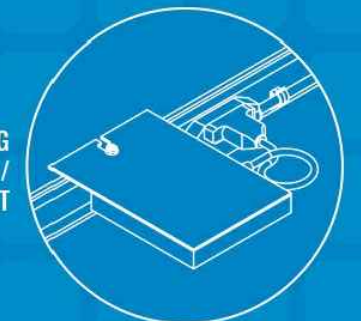
INTEGRATED BONDING
MIDCLAMP



INTEGRATED BONDING
SPLICE BAR



INTEGRATED BONDING
L-FOOT w/ T-BOLT



INTEGRATED BONDING
MICROINVERTER MOUNT w/
WIRE MANAGEMENT

 **LISTED UL2703** BONDING & GROUNDING
MECHANICAL LOADING
SYSTEM FIRE CLASSIFICATION

UNIRAC CUSTOMER SERVICE MEANS THE HIGHEST LEVEL OF PRODUCT SUPPORT



UNMATCHED
EXPERIENCE



CERTIFIED
QUALITY



ENGINEERING
EXCELLENCE



BANKABLE
WARRANTY



DESIGN
TOOLS



PERMIT
DOCUMENTATION

TECHNICAL SUPPORT

Unirac's technical support team is dedicated to answering questions & addressing issues in real time. An online library of documents including engineering reports, stamped letters and technical data sheets greatly simplifies your permitting and project planning process.

CERTIFIED QUALITY PROVIDER

Unirac is the only PV mounting vendor with ISO certifications for 9001:2015, 14001:2015 and OHSAS 18001:2007, which means we deliver the highest standards for fit, form, and function. These certifications demonstrate our excellence and commitment to first class business practices.

BANKABLE WARRANTY

Don't leave your project to chance. Unirac has the financial strength to back our products and reduce your risk. Have peace of mind knowing you are receiving products of exceptional quality. SOLARMOUNT is covered by a twenty five (25) year limited product warranty and a five (5) year limited finish warranty.

PROTECT YOUR REPUTATION WITH QUALITY RACKING SOLUTIONS BACKED BY ENGINEERING EXCELLENCE AND A SUPERIOR SUPPLY CHAIN

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